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Muhammad Ahmad

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in LLM application architecture, multi-agent orchestration systems, and knowledge graph construction. At VaconAI, built autonomous software development platforms that coordinate 6 specialized LLM agents to generate complete applications from natural language, designed triple-database knowledge extraction pipelines for legal deposition analysis, and engineered adversarial content refinement systems with blind evaluation design. Strong production experience with Claude API ecosystem, AWS Bedrock, vector databases, and async-based Python systems.

SKILLS

AI/LLM: Claude API, Claude Agent SDK, Claude Code CLI, AWS Bedrock, OpenAI API, LangChain, MCP (Model Context Protocol), Multi-Agent Orchestration, RAG, Vector Search, Knowledge Graphs

Languages: Python (Advanced), TypeScript, SQL

Backend: FastAPI, asyncio, SQLAlchemy, PostgreSQL, SQLite, Neo4j, Pinecone, MongoDB

Prompt Engineering: Multi-phase pipelines, adversarial evaluation loops, blind scoring, model routing policies

Infrastructure: AWS (Bedrock, EC2, S3), Modal.com, Docker, Git

PROFESSIONAL EXPERIENCE

AI/ML Engineer | VaconAI

Remote

Sept 2025 – Present

- Architected multi-agent autonomous development orchestrator across 5 iterations, coordinating 6 specialized LLM agents (researcher, architect, backend, frontend, tester, fixer) with full context passing; migrated from subprocess-based CLI to Claude Agent SDK, reducing codebase from 2,000+ lines to ~500 lines
- Designed intelligent decision loop using Claude Opus that dynamically routes tasks based on test results – spawns fixer agents on failure, re-runs tests until completion – eliminating manual debugging loops entirely
- Built legal deposition knowledge graph system: triple-database architecture (Neo4j + Pinecone + MongoDB) with LLM-powered extraction identifying 7 node types, 10+ relationship types with importance scoring, and async processing with semaphore-controlled concurrency
- Engineered adversarial content refinement system (creator-critic loop): Claude Sonnet creates, Opus critiques with blind evaluation, 4-dimension scoring gate (avg ≥ 7.0 AND min ≥ 5.0), checkpoint/resume, and stall detection
- Implemented model cost optimization policy across all LLM systems: Opus for orchestration/architecture decisions, Sonnet for implementation, Haiku for log condensation
- Optimized batch data processing pipeline achieving 15 $\times$ performance improvement through parallel async operations and intelligent rate limiting
- Designed and deployed serverless ML inference pipeline on Modal.com with automatic scaling; built real-time inventory analytics dashboard (Streamlit + SQL aggregation) on GCP

Software Developer | Techomnis

Remote

June 2024 – Sept 2025

- Built full-stack medical transport management system processing 500+ daily trips with reporting dashboard, CSV bulk import pipeline with validation, automated analytics reporting, and data export workflows
- Developed entity resolution model combining rule-based normalization with ML-based matching for client data deduplication with confidence scoring metrics
- Implemented backend REST APIs with advanced filtering, pagination, and aggregation for client management; delivered private production deployment with invite-only access control

SELECTED PROJECTS

IdeaRadar | Automated SaaS Opportunity Discovery Pipeline

Python, Claude API, Reddit/Product Hunt/GitHub/Kaggle APIs

VaconAI

- Scraped 5+ data sources (Reddit, Product Hunt, GitHub, 80K+ Kaggle job postings) with 5 parallel AI analysis strategies: job mining, expensive tool cloning, regional gaps, community pain points, marketplace gaps
- 100-point scoring rubric (Validation 35, Market 25, Economics 20, Execution 20) with digital viability filtering (8 SaaS feasibility criteria); MD5-based batch caching for resumable analysis across multi-hour runs

RAG-MCP | Multi-Source Research MCP Server

TypeScript, MCP SDK, Reddit OAuth, HN Algolia, X v2 API, YouTube InnerTube

VaconAI

- Exposed 11 research tools to Claude Code (HN, Reddit, X/Twitter, YouTube, web search) with local-first caching strategy, dual JSON/markdown storage, and 3,900+ cached research files

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- YouTube transcript extraction via reverse-engineered InnerTube API (no API key required); Reddit OAuth with token caching; Python bulk ingestion scripts for full channel harvesting

EDUCATION

Punjab University College of Information Technology (PUCIT)

2020 – 2024

- B.S. Computer Science
- Key Courses: Machine Learning, Data Structures, Database Systems, Statistics, Computer Vision

ACTIVITIES

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- Research Paper: Comparative Analysis of Hand Gesture Recognition Models (2024)